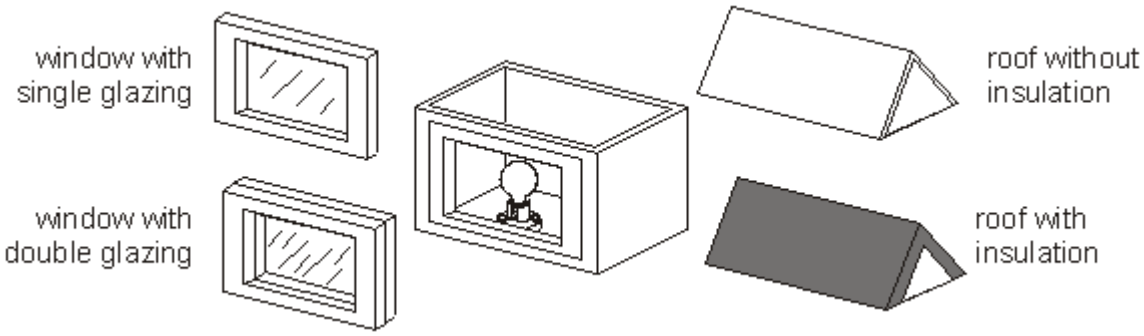
Q1.

Some pupils investigate whether double glazing or roof insulation is more efficient at reducing heat loss from houses.

They have a model house which can have these features:

- window with single glazing
- window with double glazing
- roof without insulation
- roof with insulation.



(a) A temperature sensor and a small lamp are placed inside the house. The lamp is used as a heat source.
When the model house reaches a given temperature, **the lamp is switched off**.
A datalogger then records temperature regularly over time.

(i) What can the combination of single glazing and **no** roof insulation tell pupils that is relevant to their investigation?

.....

.....

1 mark

(ii) Which **two** combinations **must** they use to find the more efficient way of preventing heat loss in their model house?

..... and .....

..... and .....

1 mark

(b) The pupils predicted that the roof insulation will be more effective than double glazing at reducing heat loss.

What evidence would support this prediction?

.....

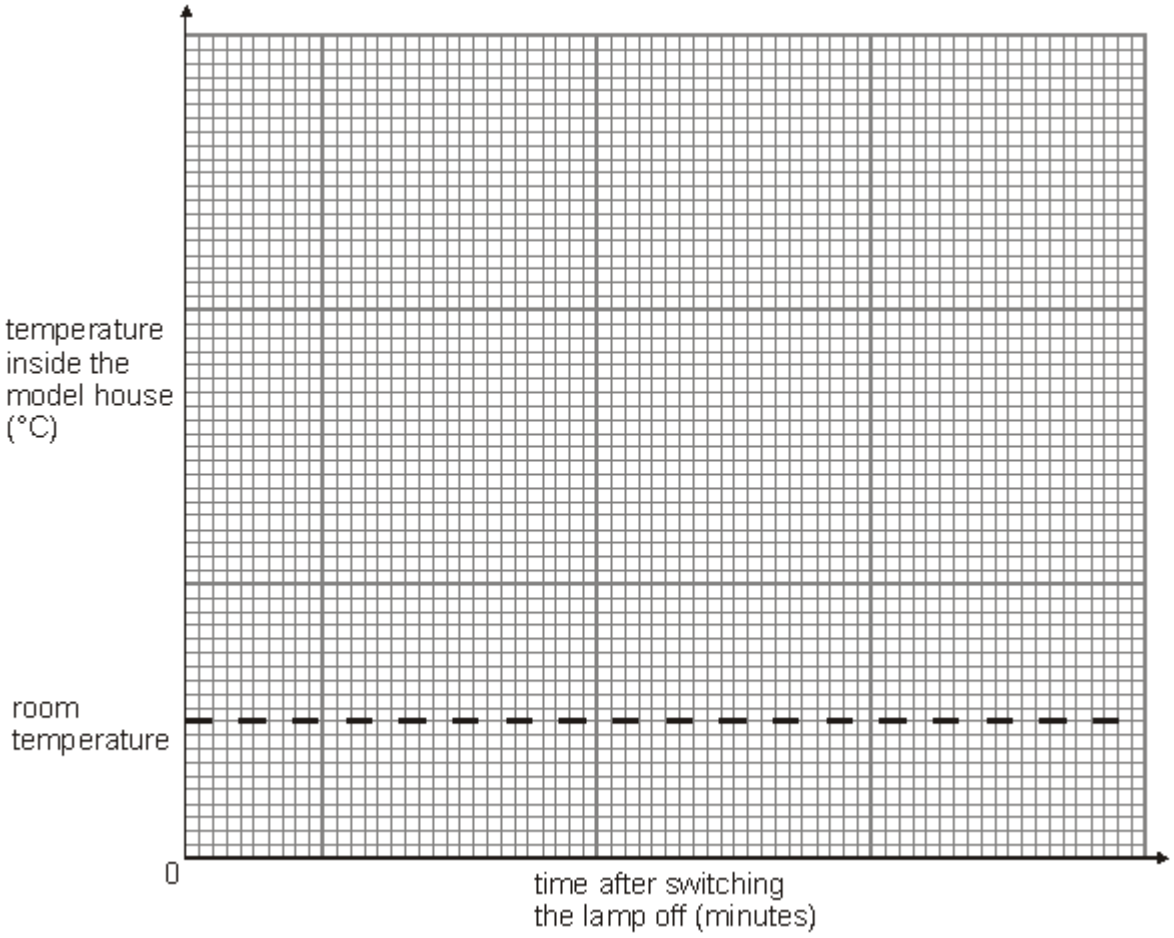
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.....

1 mark

(c) On the grid below, sketch the shape of the two graphs you would expect to see on the datalogger if the pupils' prediction is correct.

You do **not** need to add scales to the axes.
Use a solid line (_____) to show the graph for double glazed windows.
Use a dotted line (-------) to show the graph for roof insulation.



2 marks
maximum 5 marks

Q1.

(a) (i) any **one** from

- the effect of adding no extra insulation to the windows and roof
  accept 'effect of not adding insulation'
- it provides a baseline measure
- the temperature without any effort to reduce the heat loss
  *accept 'as a comparison **or** control'*
  accept 'how much energy is normally lost'
  accept 'how much heat single glazing keeps in'
  accept 'what would happen if we did nothing'

1 (L7)

(ii) double glazing *and* roof without insulation **and** a roof with insulation *and* single glazing
answers may be in either order
***both** answers are required for the mark*

1 (L7)

(b) any **one** from

- a longer time (roof insulation) to fall between two temperatures
  accept 'slower heat loss for insulation'
  accept 'faster heat loss for double glazing'
- smaller temperature fall (roof insulation) in a given time
  accept 'more heat loss for double glazing'
  accept 'roof insulation stays warmer than double glazing'
  accept 'if it is quicker for the house to reach a given temperature'

1 (L7)

(c)
- both lines **or** curves starting from the same temperature, descending and approaching (not crossing) the room temperature line

1 (L7)

- the solid line **or** curve descending more steeply than the other
  allow a difference of 5 small squares at the start
  accept correctly labelled curves which do not use solid and dotted lines as required by the question

1 (L7)

[5]